1 Multivariable Functions

1.1 Basic Definitions

In the last unit we studied functions with one input and multiple outputs. In this unit, we study functions with multiple inputs and one output. That is, we study functions of the form

$$f : D \subseteq \mathbb{R}^n \to \mathbb{R},$$

for $n \geq 2$ a natural number. We will refer to such a function as a *multivariable function*. The set $D \subseteq \mathbb{R}^n$ is the *domain* of the function, and $\mathbb{R}$ is the *codomain* of the function, which is the set in which the function takes values. One can think of the codomain as the set of all *potential* outputs of the function. The *range* is then defined to be the set of all *actual* outputs of the function.

Definition 1.1 (Range)**.** The *range* of a multivariable function f is the set

$$\text{Range } f := \{y \in \mathbb{R} : y = f(x) \text{ for some } x \in D\}.$$

The range of $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is also frequently denoted by $f(D)$. In one-variable calculus, one generally defines a function by writing $y = f(x)$, where it is understood that the domain of the function is the largest subset of $\mathbb{R}^n$ for which it is defined (i.e., no division by zero or even roots of negative numbers). This is common practice for multivariable functions as well. For example, one writes $z = f(x,y)$ meaning that the value of the function f at $(x,y) \in \mathbb{R}^2$ is z. Similarly, $w = f(x,y,z)$, means that w is the value of f at $(x,y,z) \in \mathbb{R}^3$.

As for single-variable functions, a special role is played by the following types of functions.

Definition 1.2 (Injective, surjective)**.** (a) A multivariable function f is *injective* or *one-to-one* if, for all $x_1, x_2 \in D$, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$. The contrapositive of this definition, $f(x_1) = f(x_2) \implies x_1 = x_2$, is also frequently used.

(b) A multivariable function f is *surjective* or *onto* if, for every $y \in \mathbb{R}$ there exists at least one $x \in D$ such that $y = f(x)$.

Remark 1.3. Note that a function can be injective but not surjective, surjective but not injective, neither, or both.

Example 1.4 (Examples of Multivariable Functions)**.** (a) The volume of a right circular cylinder of radius r and height h is a function $V : U \subseteq \mathbb{R}^2 \to \mathbb{R}$ defined by $V(r,h) = \pi r^2 h$.

(b) The ideal gas law is a function giving the pressure on n moles of gas in terms of the temperature and volume:

$$P : \mathbb{R}^2 \to \mathbb{R}$$
$$P(T,V) = \frac{nRT}{V}.$$

The universal gas constant, R, is a constant independent of the gas.

(c) The distance function on $\mathbb{R}^n$ is a function $\rho : \mathbb{R}^n \to \mathbb{R}$ defined by

$$\rho(x_1, x_2, \ldots, x_n) = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Exercise 1.5. For the functions in parts (a) and (c) in the example above, find the domain and range, then determine whether the function is injective, surjective, both, or neither.

Proof. (a) Since lengths are positive, the domain is the first quadrant in $\mathbb{R}^2$:

$$\text{dom } V = \{(r,h) \in \mathbb{R}^2 : r > 0 \text{ and } h > 0\}.$$

If $r > 0$ and $h > 0$, then $\pi r^2 h > 0$, so Range $V \subseteq (0,\infty)$. The function V is therefore not surjective. For any $V_0 \in (0,\infty)$, $V_0 = V(\frac{1}{\sqrt{\pi}}, V_0)$, which shows $(0,\infty) \subseteq$ Range V, and therefore Range $V = (0,\infty)$. Since $V(1,1) = V(2,\frac{1}{4}) = \pi$, V is not injective.

(c) The domain is all of $\mathbb{R}^n$. Since $\sqrt{\text{anything}} \geq 0$, Range $\rho \subseteq [0,\infty)$. The function ρ is therefore not surjective. For any $d_0 \geq 0$, $d_0 = \rho(d_0, 0, \ldots, 0)$, so $[0,\infty) \subseteq$ Range ρ, and therefore Range $\rho = [0,\infty)$. Since $\rho(1,0,\ldots,0) = \rho(0,1,0,\ldots,0)$, ρ is not injective. □

1.2 Regions in $\mathbb{R}^n$

In the study of single-variable functions, open and closed intervals played important roles. We now introduce terminology to describe important regions of $\mathbb{R}^n$.

Definition 1.6 (Open Ball, Closed Ball)**.** Let $\mathbf{a} \in \mathbb{R}^n$ and let $r > 0$. The *open ball of radius r centered at* $\mathbf{a}$ is the set

$$B_r(\mathbf{a}) = \{\mathbf{a} \in \mathbb{R}^n : ||\mathbf{x} - \mathbf{a}|| < r\}.$$

Similarly, the *closed ball of radius r centered at* $\mathbf{a}$ is the set

$$\overline{B_r(\mathbf{a})} = \{\mathbf{a} \in \mathbb{R}^n : ||\mathbf{x} - \mathbf{a}|| < r\}.$$

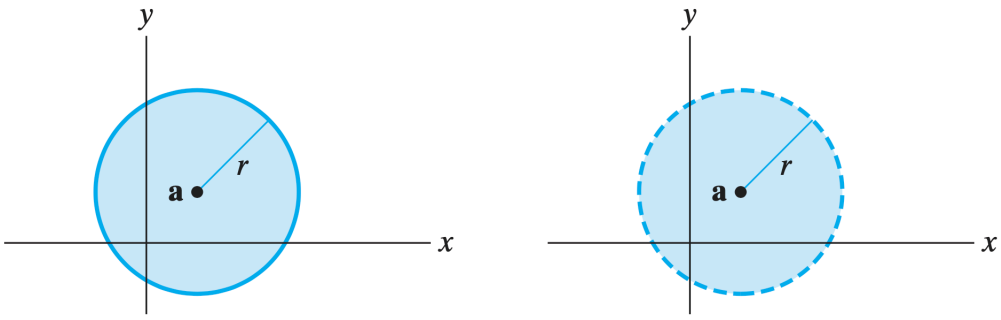


Figure 1: The closed and open balls (disks) in $\mathbb{R}^2$ defined by $||\mathbf{x} - \mathbf{a}|| \leq r$ and $||\mathbf{x} - \mathbf{a}|| < r$.

Note that, in $\mathbb{R}$, the open ball $B_r(a)$ is the open interval $(a-r, a+r)$ and the closed ball $\overline{B_r(a)}$ is the closed interval $[a-r, a+r]$.

Definition 1.7 (Interior and boundary)**.** Let $U \subset \mathbb{R}^n$ and let $\mathbf{a} \in U$.

(1) A point $\mathbf{a}$ is an *interior point* of U if there exists $r > 0$ such that $B_r(\mathbf{a}) \subseteq U$. That is, if there exists an open ball centered at $\mathbf{a}$ contained entirely in U. The *interior* of U, denoted Int U, is the set of interior points of U.

(2) A point $\mathbf{a}$ is a *boundary point* of U if for every $r > 0$, $B_r(\mathbf{a})$ contains a point of U as well as a point of $\mathbb{R}^n - U$. The boundary itself need not belong to U. The *boundary* of U, denoted Bd U or ∂U, is the set of boundary points of U.



Figure 2: Interior and boundary points.

Definition 1.8 (Open and closed sets)**.**

(1) A subset $U \subseteq \mathbb{R}^n$ is *open* if every point of U is an interior point.

(2) A subset $U \subseteq \mathbb{R}^n$ is *closed* if it contains all of its boundary points.

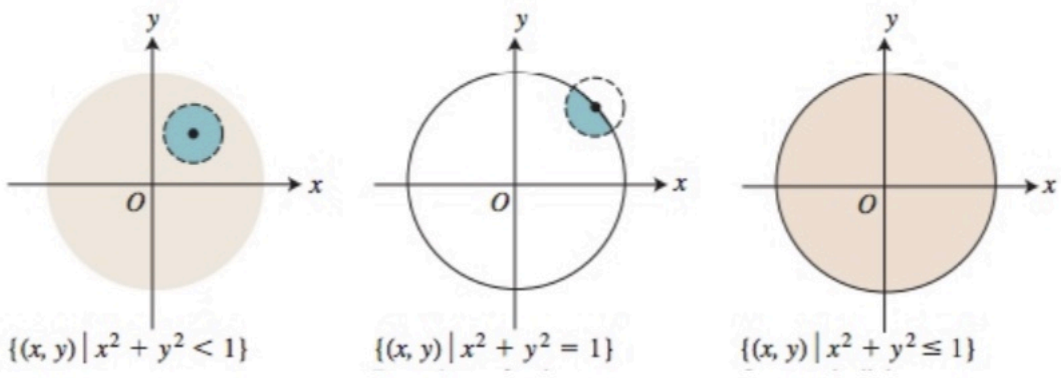


Figure 3: The boundary of the unit disc centered at the origin is the unit circle.

Example 1.9 (Examples of open sets)**.** (a) $\mathbb{R}^n$ and $\emptyset$ are open sets

(b) Every open ball is an open set

Note that a set can be neither open nor closed, or can be both open and closed.

Example 1.10. (a) We have seen above that $\mathbb{R}^n$ is open. Since $\mathbb{R}^n$ has no boundary, it trivially contains all of its boundary points. Hence, $\mathbb{R}^n$ is also closed.

(b) $(0,1] \subset \mathbb{R}$ has boundary $\{0,1\}$. It is not closed since it does not contain 0. It is not open, since any open ball centered at 1 contains points in the complement of $(0,1]$. Hence, $(0,1]$ is neither open nor closed.

Definition 1.11 (Bounded set)**.** A subset $U \subseteq \mathbb{R}^n$ is *bounded* if it is contained in some ball of finite radius. If U is not bounded, we say it is *unbounded*.

Example 1.12. (a) An open ball is bounded.

(b) A line in $\mathbb{R}^n$ is unbounded.

Exercise 1.13. Describe the domain for each of the functions in parts (a) and (c) of Example 1.4.

Proof. (a) The domain of the volume function is the first quadrant in $\mathbb{R}^2$, which is open and unbounded.
(c) The domain of the distance function is $\mathbb{R}^n$, which is both open and closed, and is unbounded. □

Example 1.14. Consider the function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) = \sqrt{y-x^2}$. For the square root to be defined, we must have $y - x^2 \geq 0$. Taking the domain D to be the largest set where the function is defined, we have then $D = \{(x,y) \in \mathbb{R}^2 : y \geq x^2\}$, which is the shaded region above (and including) the parabola $y = x^2$ shown below. Note that D is closed (since it contains its boundary $y = x^2$) and unbounded.

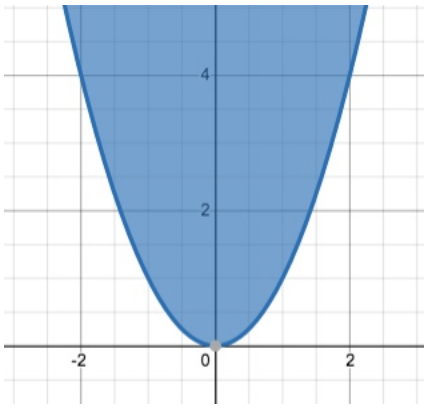


Figure 4: The domain of the function $z = \sqrt{y-x^2}$.

Exercise 1.15. Describe the domain D of the function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) = \frac{1}{\sqrt{16 - x^2 - y^2}}.$$

Proof. For the square root to be defined, we must have $16 \geq x^2 + y^2$. However, we cannot take $16 = x^2 + y^2$ since then we divide by zero. Therefore

$$D = \{(x,y) : x^2 + y^2 < 16\} = B_4(0)$$

is the open ball of radius 4 centered at $(0,0)$. Note that D is open and bounded. □

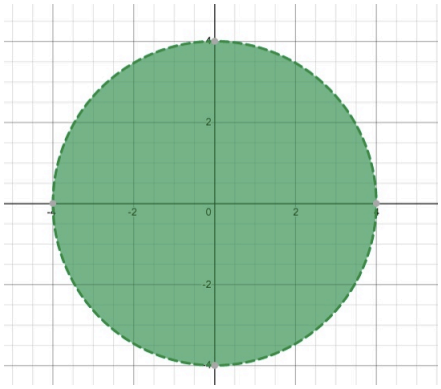


Figure 5: The domain of the function $z = \frac{1}{\sqrt{16-x^2-y^2}}$

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Exercise 1.16. For each of the following functions, evaluate $f(3,2)$ and find and sketch the domain.

1. $f(x,y) = \frac{\sqrt{x+y+1}}{x-1}$
2. $f(x,y) = x\ln(y^2 - x)$

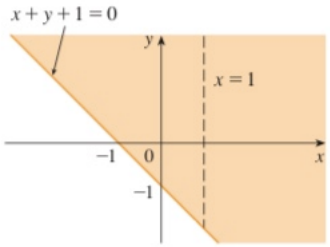
SOLUTION

(a) $$f(3,2) = \frac{\sqrt{3+2+1}}{3-1} = \frac{\sqrt{6}}{2}$$

The expression for f makes sense if the denominator is not 0 and the quantity under the square root sign is nonnegative. So the domain of f is

$$D = \{(x,y) \mid x + y + 1 \geqslant 0,\ x \neq 1\}$$

The inequality $x + y + 1 \geqslant 0$, or $y \geqslant -x - 1$, describes the points that lie on or above



the line $y = -x - 1$, while $x \neq 1$ means that the points on the line $x = 1$ must be excluded from the domain. (See Figure 2.)

(b) $$f(3, 2) = 3 \ln(2^2 - 3) = 3 \ln 1 = 0$$

Since $\ln(y^2 - x)$ is defined only when $y^2 - x > 0$, that is, $x < y^2$, the domain of f is $D = \{(x, y) \mid x < y^2\}$. This is the set of points to the left of the parabola $x = y^2$. (See Figure 3.)

Exercise 1.17. Describe the domain U of the function in Example 1.14.

Proof. It is closed (since it contains its boundary $y = x^2$) an unbounded. □

Exercise 1.18. Describe the domain U of the function in Example 1.15.

Proof. It is open and bounded. □